



THE HONG KONG
POLYTECHNIC UNIVERSITY
香港理工大學



DEPARTMENT OF
LAND SURVEYING AND GEO-INFORMATICS
土地測量及地理資訊學系

Department of land surveying and Geo-informatics

LSGI3781 Engineering Surveying



LSGI3781 Surveying Camp Report

Group [REDACTED]

GROUP MEMBER:

[REDACTED]
[REDACTED]
[REDACTED]
[REDACTED]
[REDACTED]
[REDACTED]

Supervisor:
Ms. Stella Tse

Helper: [REDACTED], [REDACTED] & [REDACTED]

Content

Abstract.....	3
What we learned	3
Summary	3
Work Allocation.....	5
Equipment List.....	6
Work Schedule	7
Control Survey (both Horizontal and Vertical)	8
Traverse (Task 1)	8
The factor of traverse error	11
Travers station summary	11
Leveling (task 2)	13
Leveling Route Planning.....	14
Leveling Calibration: Two-Peg Test	14
Leveling practical.....	14
Instrument error in leveling	17
Setting out survey and checking survey (task 3)	17
Setting-out survey Procedures	18
Radiation.....	18
Challenges faced	23
As observers.....	23
As field assistants.....	24
As bookers	24
Reflection.....	25
Conclusion	26

Abstract

The report mainly discusses the process of the measurement survey and some basic theories behind the surveying in the Tai Tong Barbecue Area Site No. In the surveying camp, 3 tasks need to be conducted, includes alignment computation, control survey (both Horizontal and Vertical) as well as setting out survey.

What we learned

	What we learned in survey camp
1	We gained practical experience and applied relevant theoretical information in the field. In this important learning opportunity, we learned about data analysis and investigative methods.
2	Since we need to work together during the measurement process, it helps us develop teamwork. Students learned about working with others, effective communication, and solving problems together.
3	In terms of the surveying process, we have encountered some difficulties during fieldwork, which require us to use critical thinking and problem-solving skills. In this process, we have developed our creativity and ability to adapt to their environment.
4	Improve professional surveying skills: The survey camp necessitated the completion of transverse, leveling, and setting out surveys within a compressed five-day timeframe, thereby cultivating our adeptness and proficiency in conducting surveys with efficiency and professionalism.
5	Our leadership proficiencies were strengthened by assuming leadership roles and actively participating in group responsibilities. For instance, as a team leader, it is crucial to possess a comprehensive understanding of the workload and allocate tasks equitably among team members. This approach fosters the development of critical decision-making and other fundamental leadership competencies.

Summary

Three major tasks were done in the survey region, the Tai Tong Barbecue region, and Site No. 1 at the Po Leung Kuk Jockey Club Tai Tong Holiday Camp. According to the Engineering Survey camp brief, our team had to finish all three critical tasks:

	Task	Detail
--	------	--------

1	Alignment Computation	Prepare the coordinates of the points to be set out
2.	Control Survey (both horizontal and vertical)	• i.e. Traversing and Levelling
3.	Checking the Radiation Setout Points	• Setting out & Checking Survey • Outlining 6 key points

On the first day of the survey camp, we held a briefing session about the site and the RO & benchmark point. A site reconnaissance and route planning process for the fieldwork was completed. The important duties mentioned above are going to be completed throughout the next three days.

To prepare the 6 initial coordinates of the points for the setting out survey, the points along the design alignment needed to be laid out on the ground were calculated based on their coordinates (Full computation in "alignment computations" sheet) The coordinates of the assigned chainages are as follows:

Chainage	Northing	Easting
CH00+08	830017.477	821201.322
CH00+16	830014.1637	821208.6025
CH00+24	830010.4457	821215.6848
CH00+32	830006.3349	821222.5466
CH00+40	830001.8442	821229.166
CH00+48	829997.2126	821235.6022

We conducted a control survey on the second and third day. Traversing and leveling were used to establish horizontal and vertical control over the alignment. This served as the reference frame for determining the points.

For planning and checking the survey, the set-out positions were identified on the ground using surveying equipment, and the coordinates were checked for accuracy using radiation measurements, angles, and distances. Six setout spots were physically

marked on the ground using pegs and the yellow cloth. Using radiation, the setout points could also be determined. Radiation measurements taken from the control points were used to verify that the locations of the six setout points were accurate enough.

When the requisite accuracy failed to be achieved, the procedure of setting out, verifying, and rectification of the points was repeated. Therefore, the primary goals were establishing horizontal and vertical control precisely, computing setout point coordinates, laying out the points on the ground, and checking them.

Work Allocation

[Each of them shall be the observer for the setting out survey at least once]

Control Survey by Traverse (Task 1)

Group Member	Duties
██████████ ██████████	Planner
██████████ ██████████	Observer
██████████	Computation
██████████ ██████████	Setting Prism
██████████	Booker
██████████	Hammer & peg
██████████ ██████████	Traverse line map
██████████ ██████████	Traverse station summary

Benchmark Verification by Leveling (Task 2)

Group Member	Duties
██████████ ██████████	Planner
██████████ ██████████ ██████████	Observer
██████████	Computation
██████████ ██████████	Staff Holder

██████████	Leveling route map
██████████	Booker
██████████	

Setting-out survey (Task 3)

Group Member	Duties
██████████ ██████████ ██████████ ██████████ ██████████ ██████████	Observer
██████████	Computation
████████████████████	Pole Holder
██████████	Taking photo
██████████ ██████████ ██████████ ██████████ ██████████ ██████████	Hammer & peg
██████████	Setting out route map
██████████	Booker

Equipment List

Task	Equipment	Quantity
Traverse (Task 1)	Tripod	3
	Total Station	1
	prism	2
	Plastic tape	1
	Hammer & peg	1 & 10
Leveling (Task 2)	Automatic Level	1
	Tripod	1
	Leveling Staff	2
	Hand Bubble	2
	Ordinary level	2
Setting-out survey (Task 3)	Tripod	1
	Total Station	1

	Hand Bubble	2
	Poler	2
	Hammer & peg	1 & 6

Work Schedule

Engineering surveying was arranged from 8 to 13 May 20██, and 3 tasks were scheduled at different times.

Day	Period	Work
1	10:45-13:00	Setting up equipment
	14:00-17:00	Site reconnaissance and Planning/Field Work and report
2	9:50-14:00	Task 1-Traverse
	14:30-18:30	Task 2-Leveling
3	9:50-18:30	Task 3 – Setting-out survey
4	9:00-13:00	Task 2-leveling

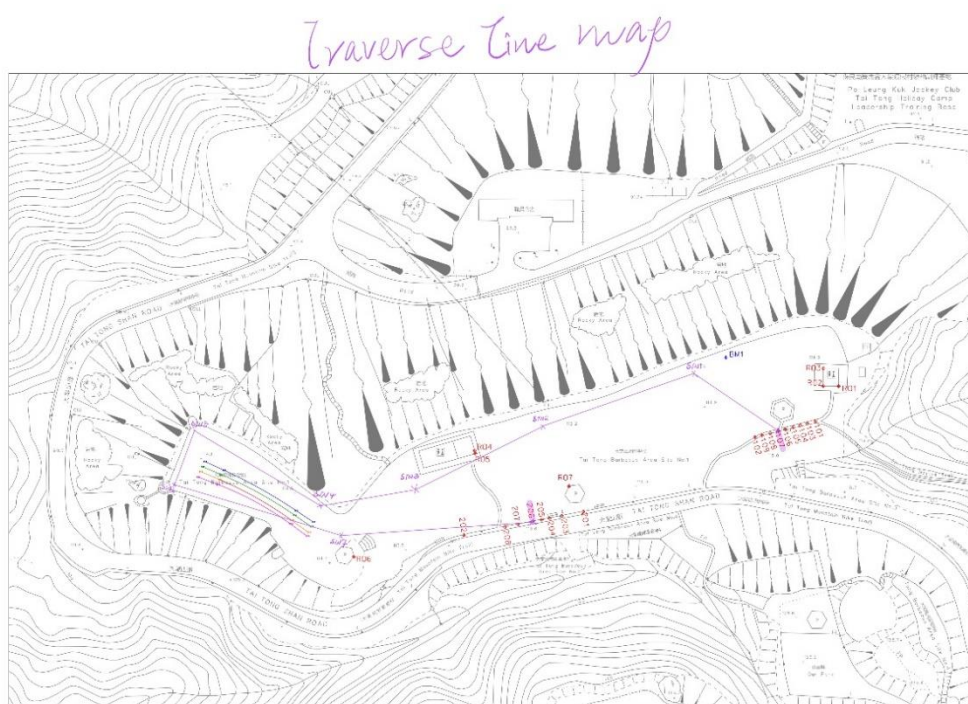
Control Survey (both Horizontal and Vertical)

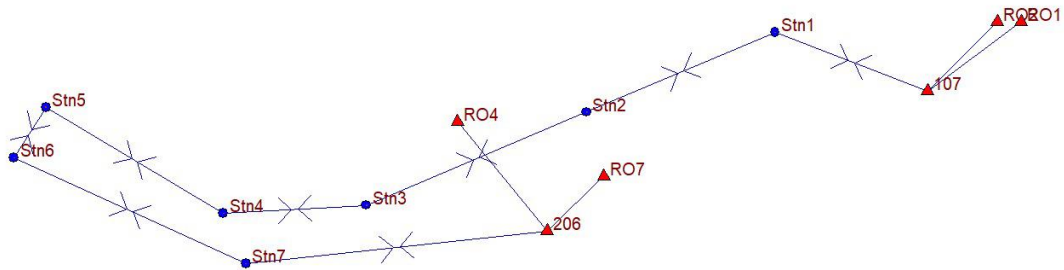
Traverse (Task 1)

The traverse survey method measures a series of lines on the ground to find horizontal and vertical coordinates of the points. During the fieldwork, previously surveyed points are used to observe the next point, so the surveyor can measure the angle and distance between each point using the total station. The resulting data is then processed and analyzed to determine the coordinates as well as the angle and other properties of each point.

In practice, the open traverse is built such that the beginning point, and the ending point are not connected. In our survey camp, we are required to conduct the transverse from 107 to 206, as shown in the maps below.

Traverse line map





Traverse field sheet

The Hong Kong Polytechnic University
Department of Land Surveying & Geo-Informatics

The Hong Kong Polytechnic University
Department of Land Surveying & Geo-Informatics

Surveying Practical Exercise

LSGI3781 Field Sheet

Date: 9/5/2019 Observer: [Redacted] Student ID: [Redacted] Group No.: [Redacted]

From	To	Face	HCR	Mean HCR	VCR	Mean VCR	Slope Distance	Horizontal Distance	Remarks
107	RO1	L	45-2-28	45-2-30	89-37-1	89-37-2			HI: 1.470
RO1	Stn1	L	53-24-19	53-24-24	89-37-9	89-37-2			
Stn1	Stn2	L	291-13-27	11-12-23	90-14-57	90-14-56	40.622	40.622	HS: 1.644
Stn2	RO4	R	111-13-27	11-12-50	269-45-4		40.622	40.621	
RO4	Stn3	R	235-25-27	53-24-24	270-22-56				
Stn3	Stn4	R	225-22-40		270-22-10				
Stn4	Stn5	L	111-13-51	11-12-53	89-36-41	89-36-41	40.620	40.620	HS: 1.558
Stn5	Stn6	L	247-1-26	247-1-26	91-5-29	91-5-26	50.878	50.878	HS: 1.458
Stn6	Stn7	R	67-1-25		268-54-37		50.878	50.878	HS: 1.563
Stn7	RO7	R	241-13-52		270-23-19		40.621	40.620	
RO7	Stn1	L	67-1-28	67-1-26	88-54-28	88-54-26	50.887	50.878	HI: 1.559
Stn1	Stn2	L	247-2-57	247-2-47	90-16-5	90-16-2	59.432	59.432	
Stn2	Stn3	R	67-2-47		269-44-0		59.432	59.432	
Stn3	Stn4	L	247-1-13		271-5-37		50.887	50.878	
Stn4	Stn5	L	67-3-0	67-3-2	89-43-83	89-43-41	59.432	59.432	HI: 1.479
Stn5	Stn6	L	266-48-3	266-48-3	87-48-46	87-48-46	35.617	35.591	HS: 1.552
Stn6	Stn7	R	86-48-30		272-11-16		35.617	35.591	
Stn7	Stn1	R	247-3-3		270-16-21		59.432	59.431	
Stn1	Stn2	L	86-48-31	86-48-30	90-16-5	90-16-2	35.616	35.590	HI: 1.539
Stn2	Stn3	L	200-53-2	200-53-2	90-16-5	90-16-2	51.398	51.392	
Stn3	Stn4	R	120-53-22		267-10-48		51.398	51.393	
Stn4	Stn5	R	200-48-14		267-81-0		35.616	35.591	
Stn5	Stn6	L	120-53-21	120-53-22	89-43-19		51.399	51.392	HI: 1.603
Stn6	Stn7	L	212-8-44	212-8-42	89-31-47		14.825	14.824	
Stn7	Stn1	R	32-8-40		270-28-15		14.824	14.824	
Stn1	Stn2	R	200-53-22		270-52-46		51.399	51.393	HI: 1.501
Stn2	Stn3	L	32-8-42	32-8-42	90-59-13		14.824	14.823	
Stn3	Stn4	L	14-30-10	14-30-10	89-3-26		63.308	63.300	
Stn4	Stn5	R	244-30-11		270-5-30		63.308	63.300	
Stn5	Stn6	R	212-8-42		269-30-65		14.823	14.823	
Stn6	Stn7	L	244-30-10	244-30-10	90-55-7		63.301	63.298	HI: 1.579
Stn7	Stn1	R	93-9-4	83-52-5	90-55-5	90-55-5	15.290	15.286	
Stn1	Stn2	R	201-52-6		269-26-14		75.290	75.286	
Stn2	Stn3	R	114-32-8		269-4-57		63.300	63.298	
Stn3	Stn4	L	263-52-5	263-52-5	89-24-17		75.290	75.286	
Stn4	Stn5	L	220-47-9	220-47-7	90-10-5				
Stn5	Stn6	L	45-53-5	45-53-5	83-38-7				
Stn6	Stn7	R	140-41-5		271-51-52				
Stn7	Stn1	R	83-52-5		269-17-56				
Stn1	Stn2	R			270-35-44				

75.290 75.285

Following the traverse survey, the data were calculated by groupmates (see traverse

Allowable Angular Misclosure $\pm 30\sqrt{n}$, n = number of setup
 Allowable Linear Misclosure: $\pm (10 + 2S/15)$, S
 = total length of traverse in metres

Traverse table

Measuring dir		HCR				Oriental		Adjusted			Control points coordinates	
From	To	D	M	S	decimal	Correction (sec)	Bearing (deg)	Adj. (sec)	WCB (deg)	WCB (dms)	Northing	Easting
107	RO2	45	2	30	45.042	-1			45.04126	45°02'29"	830035.886	821416.716
	RO1	53	24	24	53.407	0			53.40656	53°24'24"	830053.233	821434.088
	Stn1	291	13	50	291.231	0	291.23056	10	291.23333	291°14'00"	830053.183	821440.012
Stn1	107	111	13	52	111.231	-2	111.23056					
Stn2	Stn2	247	1	26	247.024	-2	247.02333	19	247.02861	247°01'43"		
	Stn1	67	1	26	67.024	-2	67.02333					
Stn3	Stn3	247	2	59	247.050	-2	247.04917	29	247.05722	247°03'26"		
	Stn2	67	3	2	67.051	-5	67.04917					
Stn4	Stn4	266	48	30	266.808	-5	266.80694	39	266.81778	266°49'04"		
	Stn3	86	48	32	86.809	-7	86.80694					
Stn5	Stn5	300	53	21	300.889	-7	300.88722	48	300.90056	300°54'02"		
	Stn4	120	53	22	120.889	-8	120.88722					
Stn6	Stn6	212	8	42	212.145	-8	212.14278	58	212.15889	212°09'32"		
	Stn5	32	8	42	32.145	-8	32.14278					
Stn7	Stn7	114	32	10	114.536	-8	114.53389	68	114.55278	114°33'10"		
	Stn6	294	32	10	294.536	-8	294.53389					
206	206	83	52	5	83.868	-8	83.86583	78	83.88750	83°53'15"	830001.129	821322.175
	Stn7	263	52	5	263.868	-8	263.86583					
	RO4	320	47	7	320.785	-8	320.78306	111	320.81387	320°48'50"	830028.420	821299.928
	RO7	45	53	51	45.898	-8	45.89528	44	45.90742	45°54'27"	830014.799	821336.285

Angular misclosure (sec)

77.5

No. of stations

9

Allowable misclosure (sec) ±90

Adjustments (sec)

9.6875

291.23333

19.375

247.02861

29.0625

247.05722

38.75

266.81778

48.4375

300.90056

58.125

212.15889

67.8125

114.55278

77.5

83.88750

Measuring direction		Adjusted bearing			Unadjusted		Corrections		Adjusted	
From	To	decimal	dms	Horiz. Dist.	N	E	N	E	N	E
									830035.886	821416.716
107	Stn1	291.23333	291°14'00	40.6115	830050.594	821378.861	-0.00284	-0.002762	830050.591	821378.859
Stn1	Stn2	247.02861	247°01'43	50.8780	830030.738	821332.018	-0.005679	-0.005525	830030.732	821332.013
Stn2	Stn3	247.05722	247°03'26	59.4320	830007.571	821277.288	-0.008519	-0.008287	830007.562	821277.279
Stn3	Stn4	266.81778	266°49'04	35.5905	830005.595	821241.752	-0.011359	-0.011049	830005.584	821241.741
Stn4	Stn5	300.90056	300°54'02	51.3925	830031.988	821197.654	-0.014199	-0.013812	830031.973	821197.640
Stn5	Stn6	212.15889	212°09'32	14.8240	830019.438	821189.764	-0.017038	-0.016574	830019.421	821189.747
Stn6	Stn7	114.55278	114°33'10	63.2990	829993.135	821247.339	-0.019878	-0.019336	829993.115	821247.320
Stn7	206	83.88750	83°53'15	75.2860	830001.152	821322.197	-0.022718	-0.022099	830001.129	821322.175
No. of stations		9						Misclosure:	-0.023	-0.022
N misclosure		-0.022718								
E misclosure		-0.022099								
Linear misclosure (m)		0.031693								
Total distance		391.3135								
Relative accuracy (1:x)		12346.91416								
Allowable misclosure		±0.062175								
Corrections										
N	E									
-0.00284	-0.002762									
-0.005679	-0.005525									
-0.008519	-0.008287									
-0.011359	-0.011049									
-0.014199	-0.013812									
-0.017038	-0.016574									
-0.019878	-0.019336									
-0.022718	-0.022099									

The factor of traverse error

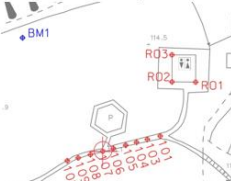
ANALYSIS OF TRAVERSE There are a few factors that might influence the reading that obtained. The factors are:

- Human factor – Readings taken on the instrument are inconsistent due to the presence of more than one observer. This error will occur to every observer. However, by following the prescribed setup and procedures, the possibility of errors can be minimized. Therefore, the observer must be the same person and double check that the reading taken is correct.
- Natural factor – Natural errors can occur due to environmental conditions. This error occurs due to wind speed, and very hot weather. Poor station selection can result in poor vision conditions due to alternating sun and shadows. The uneven ground also causes errors. This can affect our concentration and making us difficult to focus on the prism.
- Instrument factor – Instrument factors occur variations in each instrument and calibrations in each instrument. Upon acquiring results from all stations, the data was utilized to formulate a traverse adjustment report.

Travers Station Summary

The traverse station is confirmed by the computation of traverse data measurement. The summary of the traverse station was created so that it clearly demonstrated the traverse station. The following is a summary of our traverse stations:

Traverse Station Summary

TRAVERSE STATION SUMMARY	
STATION NO. 107 (FORMER NO.)	SURVEY SHEET NO.
LOCALITY	
HK 80 DATUM	HK 1980 GRID COORDINATES: N = 830035.886m, E = 821416.716m, Class
WGS 84 DATUM	Geographical Coordinates: Lat = , Long = , E, Class
Reference Frame	
STATION PHOTO	
	
LOCATION MAP (Scale 1:1000)	
	
REMARKS	
Type of Mark: Urban Survey Mark (USM)	
All measurements shown are in horizontal distance unless otherwise specified	

TRAVERSE STATION SUMMARY	
STATION NO. 208 (FORMER NO.)	SURVEY SHEET NO.
LOCALITY	
HK 80 DATUM	HK 1980 GRID COORDINATES: N = 830001.129m, E = 821322.176m, Class
WGS 84 DATUM	Geographical Coordinates: Lat = , Long = , E, Class
Reference Frame	
STATION PHOTO	
	
LOCATION MAP (Scale 1:1000)	
	
REMARKS	
Type of Mark: Urban Survey Mark (USM)	
All measurements shown are in horizontal distance unless otherwise specified	

TRAVERSE STATION SUMMARY	
STATION NO.	1 (FORMER NO.)
LOCALITY	SURVEY SHEET NO.
HK 80+ DATUM	HK 1980 GRID COORDINATES : N = 830050.591m, E = 821378.858m, Class
WGS 84 DATUM	Geographical Coordinates : Lat = N, Long = E, Class
Reference Photo	
STATION PHOTO	
LOCATION MAP (Scale 1:1000)	
REMARKS	
Type of Mark : Urban Survey Mark (USM)	
All measurements shown are in horizontal distance unless otherwise specified.	

TRAVERSE STATION SUMMARY	
STATION NO.	2 (FORMER NO.)
LOCALITY	SURVEY SHEET NO.
HK 80+ DATUM	HK 1980 GRID COORDINATES : N = 830030.732m, E = 821332.013m, Class
WGS 84 DATUM	Geographical Coordinates : Lat = N, Long = E, Class
Reference Photo	
STATION PHOTO	
LOCATION MAP (Scale 1:1000)	
REMARKS	
Type of Mark : Urban Survey Mark (USM)	
All measurements shown are in horizontal distance unless otherwise specified.	

TRAVERSE STATION SUMMARY	
STATION NO.	3 (FORMER NO.)
LOCALITY	SURVEY SHEET NO.
HK 80+ DATUM	HK 1980 GRID COORDINATES : N = 830007.562m, E = 821277.279m, Class
WGS 84 DATUM	Geographical Coordinates : Lat = N, Long = E, Class
Reference Photo	
STATION PHOTO	
LOCATION MAP (Scale 1:1000)	
REMARKS	
Type of Mark : Urban Survey Mark (USM)	
All measurements shown are in horizontal distance unless otherwise specified.	

TRAVERSE STATION SUMMARY	
STATION NO.	4 (FORMER NO.)
LOCALITY	SURVEY SHEET NO.
HK 80+ DATUM	HK 1980 GRID COORDINATES : N = 830006.584m, E = 821241.741m, Class
WGS 84 DATUM	Geographical Coordinates : Lat = N, Long = E, Class
Reference Photo	
STATION PHOTO	
LOCATION MAP (Scale 1:1000)	
REMARKS	
Type of Mark : Urban Survey Mark (USM)	
All measurements shown are in horizontal distance unless otherwise specified.	

TRAVERSE STATION SUMMARY	
STATION NO. 6 (FORMER NO.)	SURVEY SHEET NO.
LOCALITY	
HK 80 DATUM	HK 1980 GRID COORDINATES: N = 820031.973m, E = 821197.640m, Class: 1
WGS 84 DATUM	GEOGRAPHICAL COORDINATES: Lat = , Long = , E, Class: 1
Reference Plane: m above HKPD and measured to the top of mark, Class: 1	
STATION PHOTO	
LOCATION MAP (Scale 1:1000)	
REMARKS	
Type of Mark: Urban Survey Mark (USM)	
All measurements shown are in horizontal distance unless otherwise specified.	

TRAVERSE STATION SUMMARY	
STATION NO. 6 (FORMER NO.)	SURVEY SHEET NO.
LOCALITY	
HK 80 DATUM	HK 1980 GRID COORDINATES: N = 830019.421m, E = 821189.747m, Class: 1
WGS 84 DATUM	GEOGRAPHICAL COORDINATES: Lat = , Long = , E, Class: 1
Reference Plane: m above HKPD and measured to the top of mark, Class: 1	
STATION PHOTO	
LOCATION MAP (Scale 1:1000)	
REMARKS	
Type of Mark: Urban Survey Mark (USM)	
All measurements shown are in horizontal distance unless otherwise specified.	

TRAVERSE STATION SUMMARY	
STATION NO. 7 (FORMER NO.)	SURVEY SHEET NO.
LOCALITY	
HK 80 DATUM	HK 1980 GRID COORDINATES: N = 829993.115m, E = 821247.320m, Class: 1
WGS 84 DATUM	GEOGRAPHICAL COORDINATES: Lat = , Long = , E, Class: 1
Reference Plane: m above HKPD and measured to the top of mark, Class: 1	
STATION PHOTO	
LOCATION MAP (Scale 1:1000)	
REMARKS	
Type of Mark: Urban Survey Mark (USM)	
All measurements shown are in horizontal distance unless otherwise specified.	

Leveling (task 2)

Leveling is used to determine differences in elevation between points. It establishes vertical control and the vertical relationship between points. Leveling is done using an instrument called dumpy level, which measures the vertical angle between the line of sight and the horizontal plane. Repeated measurements at different stations give the

elevation difference. Leveling observations have to be recorded properly in leveling notes along with staff readings, backsight, and foresight points. The result of the leveling process captures critical information pertaining to the differences in elevation among various points.

Leveling Route Planning

The initial step in the leveling survey was to locate all benchmarks and horizontal control points. Our team successfully identified the positions of these benchmarks and control points, facilitating the optimal route planning between them. The path for the leveling route was determined based on the shortest distance between benchmarks and travel time, with the objective of minimizing instrument setups and potential errors. It was ensured that all mandatory benchmarks and control points were connected as required for the survey, leaving no point unaccounted for.

Leveling Calibration: Two-Peg Test

The Two-Peg Test was performed to verify the precision and dependability of the leveling survey's measurements. It evaluated the level's ability to display the true vertical position required for accurate leveling. The test findings confirmed that the leveling measures were accurate and precise.

The two peg test results were as follows:

	BS	FS
A1	1.388	
B1		1.381
A2	1.257	
B2		1.248

Misclosure: 0.002 m

The Two-Peg Test assessed the variance in vertical readings over a predetermined baseline distance. It determined how far the level readings deviated from the real vertical locations across the test baseline. This measured the degree of precision that could be expected from the instrument. It must be done before starting the actual leveling survey.

Leveling practical

Task 2 is leveling, and we begin by moving from the benchmark and then passing through the setting-out survey area. We employ close loop leveling with the surveying camp's recommendations, which we set 9 CP in the leveling path.

Level route map

leveling route map



scale 1 : 1000

Leveling field sheet

Station	BS	IS	FS	Rise +	Fall -	Raw RL	Adj	R.L.
A1	1.388						(mm)	
B1			1.381					
A2	1.257							
B2			1.248					
BM	1.230							114.1980
CP1	1.192		2.6129		0.899	113.299	0.11	113.2991
CP2	1.099		1.400		0.208	113.091	0.22	113.0912
CP3	2.430		0.960	0.139		113.230	0.33	113.2303
Stn4		1.425		1.005		114.235	0.33	114.2353
CP4	0.768		1.623		0.198	114.037	0.44	114.0374
Stn5		1.422			0.654	113.383	0.44	113.3834
CP5	1.452		0.605	0.817		114.200	0.56	114.2006
Stn5		2.269			0.817	113.383	0.56	113.3836
Stn4		1.417		0.852		114.235	0.56	114.2356
CP6	0.415		1.671		0.254	113.981	0.67	113.9817
CP7	1.473		1.343		0.928	113.053	0.78	113.0538
CP8	1.899		1.219	0.254		113.307	0.89	113.3079
BM			1.009	0.890		114.197	1	114.1980
	11.958		11.959					

The measurement data are documented as follows:

Point	BS	IS	FS	Rise	Fall	Unadjusted lvl.	Adj. (mm)	Reduced level
BM	1.230							114.1980
CP1	1.192		2.129	0	0.899	113.2990	0.11	113.2991
CP2	1.099		1.400	0.000	0.208	113.0910	0.22	113.0912
CP3	2.430		0.960	0.139	0	113.2300	0.33	113.2303
Stn4		1.425		1.005	0	114.2350	0.33	114.2353
CP4	0.768		1.623	0	0.198	114.0370	0.44	114.0374
Stn5		1.422		0	0.654	113.3830	0.44	113.3834
CP5	1.452		0.605	0.817	0	114.2000	0.56	114.2006
Stn5		2.269		0.000	0.817	113.3830	0.56	113.3836
Stn4		1.417		0.852	0	114.2350	0.56	114.2356
CP6	0.415		1.671	0.000	0.254	113.9810	0.67	113.9817
CP7	1.473		1.343	0	0.928	113.0530	0.78	113.0538
CP8	1.899		1.219	0.254	0	113.3070	0.89	113.3079
BM			1.009	0.890	0	114.1970	1.00	114.1980

Sum BS	11.958
Sum FS	11.959
Misclosure	0.001
No. of stations	9
Allowable misclosure	± 0.009
Adjustments (mm)	
0.11111	
0.22222	
0.33333	
0.44444	
0.55556	
0.66667	
0.77778	
0.88889	
1	

Instrument error in leveling

1	parallax	The parallax can't be completely eliminated. This mistake is caused by insufficient eyepiece and object-glass focusing. Before beginning any reading, the parallax should be eliminated through good focusing.
2	Movement of the level staff during observing	Sometimes when we are changing the leveling station, the groupmate unintentionally moves the level staff. The leveling staff has slightly shifted from its original position, resulting in an inaccurate leveling issue. As a result, our group mate must be trained to shift the staff to the correct locations.
3	Staff not held vertically.	The observer may need a long time to observe the leveling staff's height, thus testing the staff man's patience and focus. Wind and carelessness may cause a bubble to drift away from the middle circle during observation.

Setting out survey and checking survey (task 3)

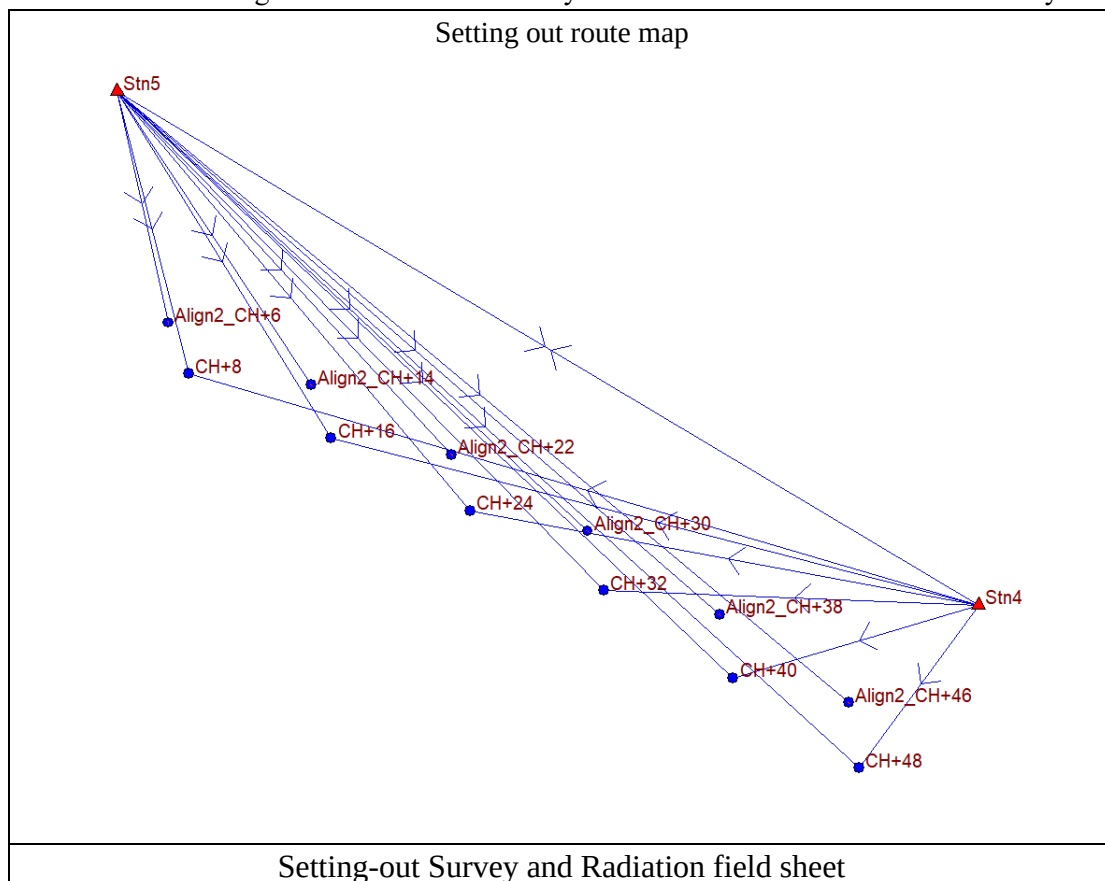
The establishment of the marks and lines to define the position and level of the elements for the construction work so that works may proceed concerning them. Setting out involves locating and marking points on the ground whose positions have been previously determined through survey computations. It involves establishing the positions of points that will be used during construction like the center line of roads, building corners, etc. The coordinates of the points to be set out are obtained from the coordinates computed from the control survey. These are used to locate the points on the ground using the total station. Furthermore, horizontal control and vertical control have to be established before setting out any points. Applying the radiation method while checking the setout points.

Setting-out survey Procedures

A setting out survey is a fundamental task in engineering and construction that involves marking out the position of structures or features on a site. It typically requires the use of a prism or target, which is a reflective object that enables the surveying equipment to measure distances and angles accurately. Before beginning the setting-out surveying, the setting-out data must be computed to acquire the horizontal and height of the design points.

Radiation

To assess the accuracy of the design points, we will do radiation and determine 6 points from the other group. After the radiation, perform the computation to compare the 6 design points between the designed coordinate and surveyed coordinate to determine data accuracy.



The Hong Kong Polytechnic University									
Department of Land Surveying & Geo-Informatics									
Surveying Practical Exercise									
LSGL3781 Field Sheet									
Date:		Observer:		Student ID:			Group No.:		
From	To	Face	HCR	Mean HCR	VCR	Mean VCR	Slope Distance	Horizontal Distance	Remarks
Stn 4									
	Stn 5	L	300-53-45		90-49-45		51.399	51.399	HT: 1.439
	CH+8	L	289-23-44		90-46-6		42.184	42.180	HS: 1.550
	CH+16	L	289-30-58		89-55-27		34.215	34.215	HS: 1.650
	CH+24	L	280-34-11		89-55-54		26.510	26.510	"
	CH+32	L	272-14-21		89-24-50		19.220	19.219	"
	CH+40	L	253-26-21		88-41-10		13.124	13.121	"
	CH+48	L	216-15-14		88-27-13		10.387	10.383	"
Stn 5									
	Stn 4	L	120-53-45		89-9-16		51.400	51.394	HT: 1.513
	CH+8	L	165-45-59		88-51-19		14.968	14.965	HS: 1.430
	CH+16	L	148-22-52		87-33-18		20.920	20.901	HS: 1.650
	CH+24	L	140-0-46		88-3-42		28.079	28.082	"
	CH+32	L	135-48-23		88-11-23		35.771	35.753	"
	CH+40	L	133-44-29		88-24-20		43.625	43.608	"
	CH+48	L	131-27-9		88-40-56		51.481	51.467	"
	Adj CH+6	L	167-24-33		86-53-22		12.115	12.158	"
	Adj CH+12	L	146-31-27		87-15-13		18.072	18.051	"
	Adj CH+18	L	131-27-26		87-56-46		25.232	25.216	"
	Adj CH+24	L	133-0-56		88-8-59		33.024	33.017	"
	Adj CH+30	L	131-8-16		88-18-30		40.918	40.900	"
	Adj CH+36	L	129-50-14		88-33-28		48.872	48.857	"

In the survey, using the given coordinate to calculate the coordinates and the distance of the 6 designed points and the computation of the 6 designed points shown below:

The computation of 6 designed points																								
		HCR				VCR				Design coordinates										Measured coordinates			Delta	
From	To	D	M	S	decimal	D	M	S	decimal	Slope Dist	HI	HS	Hor Dist	Ver Dist	R.L.	N	E	Z	N	E	Z	N	E	Z
Control survey for Alignment 3																								
Stn4										1.44					114.126				830005.58	821241.74	114.126			
	Stn5	300	53	45	300.8958	90	49	45	90.8292	51.398	1.89	51.39262	-0.74379	113.13101	830031.973	821197.640		830031.973	821197.641	113.131	0.000	0.001		
	CH+8	286	23	44	286.3956	92	46	6	92.7683	42.184	1.65	42.13477	-2.03739	111.87741	830017.48	821201.32		830017.477	821201.319	111.877	0.000	-0.003		
	CH+16	284	30	58	284.5161	89	55	27	89.9242	34.215	1.65	34.21497	0.04526	113.9608	830014.16	821208.6		830014.160	821208.616	113.960	-0.004	0.016		
	CH+24	280	34	11	280.5697	89	55	54	89.9317	26.51	1.65	26.50996	0.03162	113.94642	830010.45	821215.68		830010.446	821215.681	113.946	0.001	-0.004		
	CH+32	272	14	31	272.2419	88	9	50	88.1639	19.22	1.65	19.21013	0.61582	114.53062	830006.33	821222.55		830006.335	821222.545	114.531	0.000	-0.001		
	CH+40	253	26	21	253.4392	88	41	10	88.6861	13.124	1.65	13.12055	0.30093	114.21573	830001.84	821229.17		830001.844	821229.165	114.216	0.000	-0.001		
	CH+48	216	15	14	216.2539	88	27	13	88.4536	10.387	1.65	10.38322	0.28031	114.19511	829997.21	821235.6		829997.211	821235.601	114.195	-0.002	-0.002		

Designed point CH+8 close-up photo



CH+8 overall view photo



Designed point CH+16 close-up photo

CH+16 overall view photo



Designed point CH+24 close-up photo



Designed point CH+32 close-up photo



Designed point CH+40 close-up photo



CH+24 overall view photo



CH+32 overall view photo



CH+40 overall view photo



Designed point CH+48 close-up photo

CH+48 overall view photo



Coordinate calculation for 12 survey points

Measured coordinates of the 12 points
(Align.3 CH+8 to +48, Align.2 CH+6 to +46)

Alignment 3					
Coordinates on the curve					
Chainage	Northing	Easting			
8	830017.477	821201.322			
16	830014.164	821208.602			
24	830010.446	821215.685			
32	830006.335	821222.547			
40	830001.844	821229.166			
Coordinates on straight line					
Chainage	Northing	Easting			
48	829997.213	821235.602			
Alignment 2					
Coordinates on the curve					
Chainage	Northing	Easting			
14	830016.888	821207.591			
22	830013.319	821214.75			
30	830009.388	821221.716			
38	830005.105	821228.472			
46	830000.481	821235			
Coordinates on straight line					
Chainage	Northing	Easting			
6	830020.097	821200.264			

	830031.973	821197.640	113.383
35	830005.583	821241.742	114.225
39	830017.467	821201.320	113.546
40	830014.175	821208.598	114.139
36	830010.456	821215.687	114.197
41	830006.338	821222.563	114.377
45	830001.850	821229.172	114.460
45	829997.234	821235.615	114.430
	830020.100	821200.256	113.889
	830016.899	821207.571	114.112
	830013.321	821214.757	114.154
	830009.393	821221.728	114.313
	830005.111	821228.482	114.454
	830000.588	821235.082	114.477

Comparison of the coordinates between the surveyed points and the design point (Delta N, E, Z)			
	-0.0005193	0.0008165	-0.0104248
	-0.0096060	-0.0024619	0.0867521
	0.0110051	-0.0048259	0.0692251
	0.0105049	0.0017592	0.1407187
	0.0035104	0.0163882	0.1554543
	0.0054630	0.0057579	0.1349795
	0.0210092	0.0123492	0.1256848
	0.0029105	-0.0078374	
	0.0109687	-0.0205958	
	0.0020637	0.0070869	
	0.0050404	0.0118465	
	0.0059570	0.0099360	
	0.1067732	0.0825980	

The closer the measured positions of CH+8, CH+16, CH+24, CH+32, CH+40, and CH+48 are to their design points, the more accurate the setting-out survey. The Delta N and Delta E coordinates of the six measured points differed by a few millimeters from the design points, as expected. However, Delta Z varied from 6.9cm to 15.5cm, indicating inconsistent elevation of the radiation poles. In the control survey, the poles were on the ground, while in the check survey, they were on top of the nails, causing the Delta Z error. Although Delta Z is less critical than Delta N and Delta E, it still affects the overall accuracy. Nonetheless, all Delta N and Delta E values of our stations met the accuracy standard of less than 12mm, indicating high precision and accuracy in the setting-out survey technique.

Challenges faced

As observers

We encountered several difficulties while conducting our traversing and leveling measurements.

- (1) For traversing, we did not always accurately set the true bearing during the initial round of measurement. However, we were able to identify the orientation error after measuring both faces, and we solved the problem by carefully measuring the backsight prism multiple times to reduce error.
- (2) For leveling, the markings on the leveling staff were only accurate to the centimeter, so we had to estimate the actual reading. This led to inconsistencies in interpreting the readings and introduced potential biases.
- (3) Regarding the double intermediate sight, we were a bit confused about how to do it at first. We were surprised to find out that we need to find an intermediate point by ourselves.
- (4) For both traversing and leveling, parallax was a problem that occurred during both surveying methods. Parallax caused the image of the object to shift while observing, resulting in inconsistent measurements. This was our very first-time performing survey measurements on the grassy ground. Since we did not realize the tripod legs could sink into the unstable soil, this initially led to inconsistent measurements. Later, we learned to firmly plant the tripod legs and the change plate in the ground to ensure the station height remained stable during measurements. By taking these

corrective measures, we were able to reduce the errors and complete our traversing and leveling measurements.

As field assistants

- (5) The process of conducting a setting out survey is often challenging, particularly when multiple students are simultaneously engaging in the activity. Specifically, locating the prism or target can prove to be a daunting task due to the presence of numerous individuals in the vicinity. When several students are conducting the survey simultaneously, it can be challenging to locate the prism amidst the crowd, leading to delays and potential errors in the survey. However, it seems that there is no solution to this problem and double-check and plan carefully is the best way to conduct the setting out survey.
- (6) In order to obtain accurate results, the leveling process was conducted three times. During the initial attempt, an error was introduced due to a miscalculation in the intermediate sight. In the second attempt, the leveling staff was inadvertently placed in an incorrect position, and this error went unnoticed until it was observed by another group.
- (7) The transverse surveying process encountered challenges due to some points being located under trees, which impeded the observer's ability to obtain a clear view of the prism and, consequently, accurate results. Despite employing flashlights or seeking assistance from team members, the observer encountered considerable difficulty in sighting the prism. In this case, we shifted our points in a brighter location as our solution to this problem.

As bookers

- (8) The booker may face time constraints when recording data, potentially resulting in some data being missing or omitted. During the check survey, the sun had already set, so the booker had to quickly write down the data to save time measuring remaining poles. Instead, the booker took photos and recorded the data later.

Overall, the experience highlighted the importance of careful setup, double-checking measurements, and troubleshooting techniques to resolve issues that arise during surveying work. With more practice and proper equipment, we can improve the accuracy and efficiency of our measurements in the future.

Reflection

– Any decision(s) you have made and why?

- (1) Determining the number of measurements and instrument setups required. More measurements and setups lead to higher accuracy but also greater time and cost. The decision ought to ensure adequate precision and redundancy without being excessive. Repeating measurements at some station points also helps detect errors.
- (2) Selecting the leveling route and traverse order. Carefully planning the path to minimize the number of instrument setups can reduce errors and time/cost. However, the route should still allow for measurements between all required points. The traverse order also affects precision as errors accumulate.
- (3) Deciding on the staff reading, e.g., 1st staff, halves, quarters, or eighths. Using a smaller gradation leads to higher precision but lower accuracy due to instability and parallax errors. The reading should match the required precision and be the smallest that can be read consistently.
- (4) Setting out involves passing through six points, with each point capable of seeing and passing through the chainages. However, due to roots from nearby trees, points located under them cannot be observed, rendering the task difficult. In such situations, there is no optimal solution to overcome this challenge. To ensure successful pegging, small pegs are used in traverse, while long pegs are used in setting out. It is crucial to select points with soft soil to prevent difficulties while pegging. To cover 42 meters and several chainages ranging from CH00+08 to CH00+48, each eight meters long, plastic tape is used. Before measuring the chainages, a box-like structure is needed to fix them, and other student's assistance is necessary to ensure the tape was straight and aligned with the azimuth. Using red points, the laser beam's emission from the total station is traced to locate the accurate position on the ground, followed by the hammering process.
- (5) In traverse, the prism's position must be precisely remembered to avoid setting it in the wrong place, which can lead to time wastage. In leveling, the station should not have obstructions or significant height differences, and a gentle slope is not necessary to enhance visibility. Two intermediate stations are required, located closer to the observers, and providing a better view.

Conclusion

In conclusion, surveys are crucial in construction projects, providing essential data for design, planning, and executing work. Control surveys establish the horizontal and vertical control points for subsequent tasks like traverses and leveling.

However, potential errors during traverse surveys must be considered, like human, natural and instrument errors. While it may not be possible to eradicate errors entirely, their occurrence can be minimized through careful planning and the implementation of double-checking measures.

Leveling is particularly important for determining differences in elevation between points and establishing vertical control and relationship between points. Leveling route planning helps optimize the surveying process by minimizing potential errors and reducing the number of instrument setups required.

Instrument errors can significantly impact the accuracy of leveling measurements. The three significant sources of error in leveling are parallax, movement of the level staff during observation, and staff not held vertically. These errors can lead to discrepancies in the readings and cause significant issues in the surveying process.

Finally, setting out and checking surveys involves establishing the positions of points to be used during construction, such as building corners and center lines of roads. By following these surveying procedures and tasks, construction projects can proceed with the necessary accuracy and precision, ensuring safe and successful completion.